Student Research Symposium PROGRAM

GEORGIA SOUTHERN UNIVERSITY

ALLEN E. PAULSON COLLEGE OF ENGINEERING AND COMPUTING

April 24, 2026

Information Technology Building
1100 IT Drive, Statesboro, GA 31458
9:00 AM (Check-In) to 5:00 PM

Complimentary Event Registration: Click HERE.

PARKING: No parking pass is required when you park in the Parking Lot 41 immediately behind the building.
DO NOT PARK IN ANY PAY TO PARK SPACES

Pull into the parking lot from Forest Drive and proceed to Parking Lot 41.

PROGRAM

9:00 am - Welcome and Check-In - 1st Floor Lobby
9:30 am to 11:30 am - Undergraduate Poster Session - 1st Floor
11:30 am to 1:00 pm - Career Fair - 2nd Floor Mezzanine - Register
1:00 pm to 3:00 pm - Graduate Poster Session - 1st Floor
3:00 pm to 4:00 PM - Reception - 1st Floor Lobby
4:00 pm - Keynote Address - IT1004 Auditorium
4:30 pm to 5:00 pm - Awards Ceremony - IT1004 Auditorium

Poster Presentations

The posters are presented in 10 categories with two sessions per the schedule below:

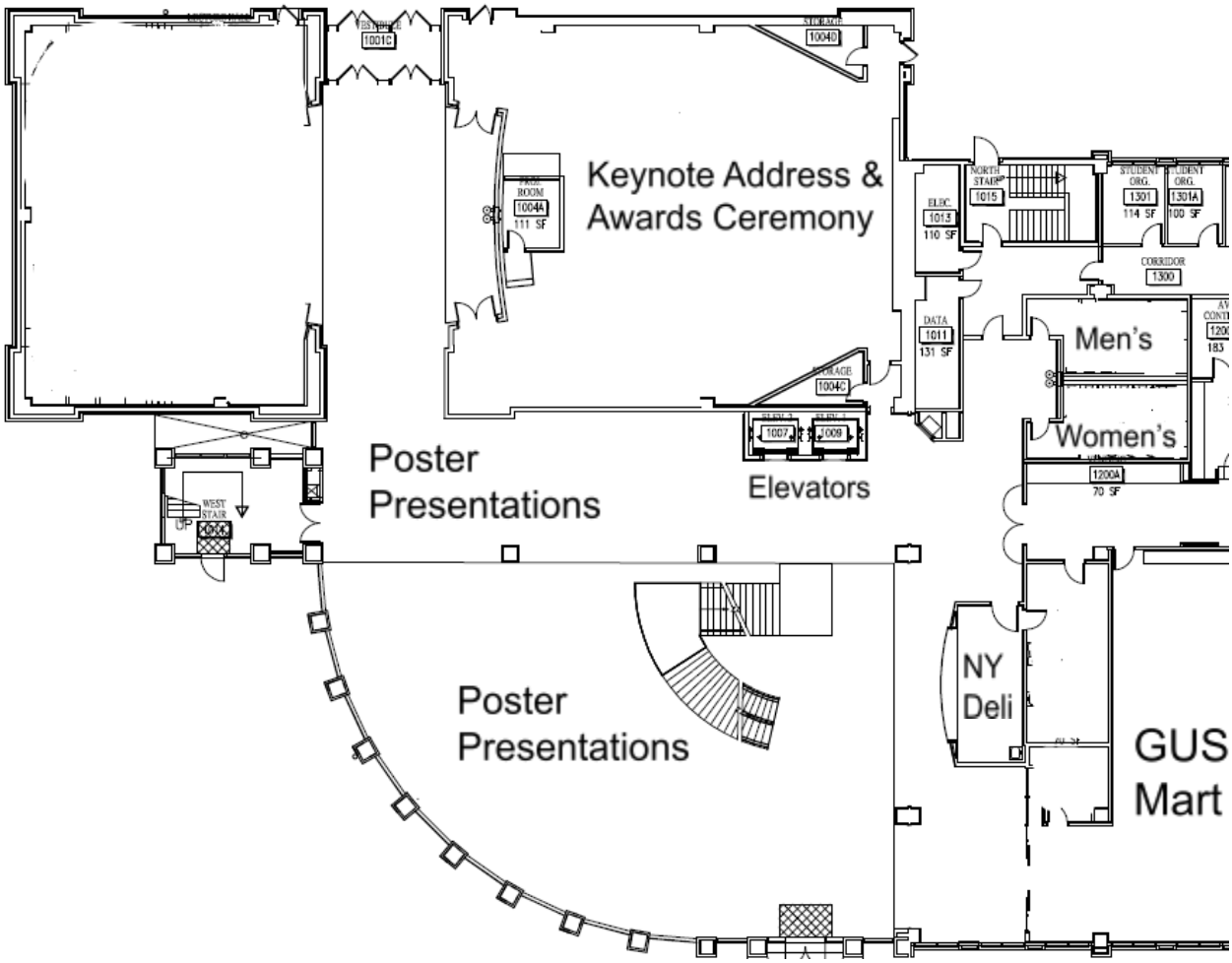
AM Session - Undergraduate: 9:30 AM to 11:30 AM
PM Session - Graduate and Mixed teams: 1:00 PM to 3:00 PM

Morning and Afternoon Poster Presentations			
Category	**Poster Range**	**Undergrad 9:30 - 11:30 AM**	**Grad 1:00 - 3:00 PM**
AI/ML, Data Science, & Predictive Analytics	1-29	8	14
Robotics & Automation	30-59	12	3
Cybersecurity & Data Protection and Networking & Communication Systems	60-79	4	7
Structural & Civil Engineering	80-99	7	7
Transportation & Automotive Technology	100-119	6	2
Environmental & Sustainability	120-139	2	6
Biomedical, Health, & Computer Interface	140-169	14	5
Manufacturing & Material Science	170-189	5	8
Power & Energy Systems	190-209	10	3
Simulation & Modeling	210-229	3	7
Totals		**71**	**62**

Judges, Guests, Career Fair participants can join us for lunch in the Professional Development Center, Room 3318. Take the lobby elevator up to the 3rd floor, exit left and and walk all the way down the hall to the east end of the building. Room 3318 is at the end of a short hallway on the left (see map on page 5).

For a full list of Categories and Posters, please see Poster Information.

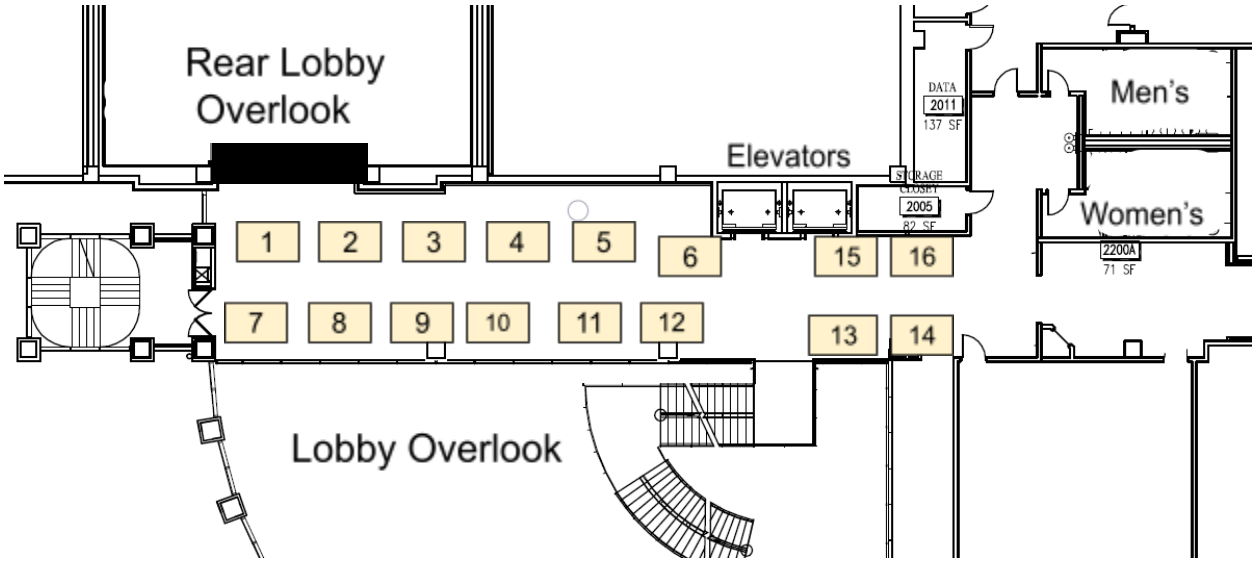
POSTER PRESENTATION LAYOUT - 1st Floor



See onsite program for location of each poster category.

CAREER FAIR - 2ND FLOOR MEZZANINE

11:30 am to 1:00 pm



The career fair features “tall-boy” tables which are perfect for casual conversations, while making the most of the space available. Students should register for this event on Handshake.

Table Assignments (All SPOTS ARE Filled)

Company	**Table**		**Company**	**Table**
Claxton Poultry Farms	1		Port City Logistics	9
Ecoplastic America	2		HMGMA	10
Crider Foods	3		Brodie International	11
Gulfstream	4		Georgia Ports	12
Thomas and Hutton	5		Southern Nuclear	13
Daniel Defense	6		Rolls-Royce	14
Mercer Engineering Research Labs	7		Brasseler-Henry Schein	15
Trane Technologies	8		KSB GIW, Inc.	16

Luncheon - 3rd Floor Room 3318

11:30 am to 1:00 pm

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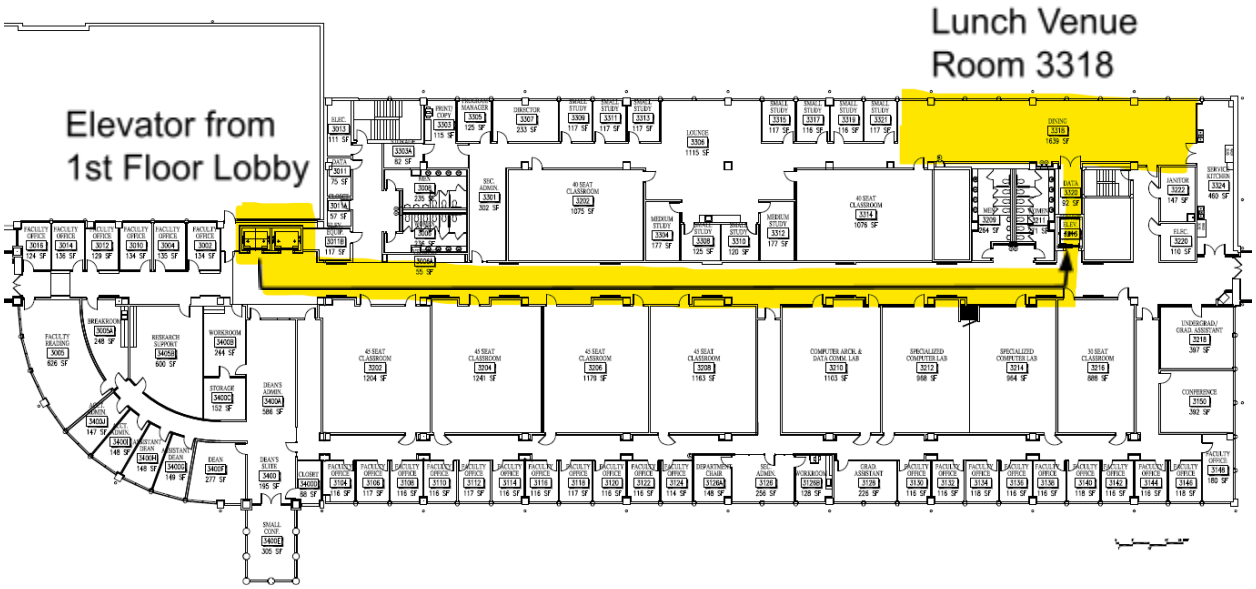

THIRD FLOOR PLAN
SCALE: NONE

Keynote Address and Awards Ceremony

1st Floor - Room 1004, Information Technology Building

4:00 pm to 5:00 pm

PROGRAM

- Welcome, Reinhold Gerbsch, Director of Industry Relations
- Greetings and Introduction of Keynote Speaker, Rami Haddad, Interim Dean
- Keynote Address, Mr. Johnny Hodges, Vice President, Engineering and Flight, Gulfstream Aerospace Corporation
- Awards Introduction, Reinhold Gerbsch
- Awards and Photographs
- Closing

Johnny Hodges
Vice President, Engineering
Gulfstream Aerospace

Johnny Hodges is Vice President of Engineering at Gulfstream Aerospace, where he provides strategic and operational leadership for the design, development, certification, and lifecycle support of Gulfstream's aircraft portfolio. In this role, he oversees multidisciplinary engineering organizations and ensures the delivery of innovative, safe, and reliable products that meet the highest standards of performance and regulatory compliance.

With more than 30 years of experience in aerospace and transportation, Johnny brings deep technical expertise combined with a proven record of executive leadership. Prior to his current position, he served as Director of Aircraft Structures, where he was responsible for structural design, aircraft loads and dynamics, materials and processes, and stress disciplines across the Engineering organization. His teams supported all Gulfstream product lines, including advanced research initiatives and new aircraft development programs.

Since joining Gulfstream in 1996, Johnny has held a series of roles with increasing responsibility across engineering, engineering leadership, preliminary design, and operations support. His career reflects a sustained commitment to engineering excellence, innovation, and the development of high-performing teams in complex, highly regulated environments.

Johnny holds a Bachelor of Science degree in Mechanical Engineering from Mercer University.

AWARD SPONSORS

The Student Research Symposium awards are made possible through the generous support of our industry partners and sponsors. Their commitment to advancing education, innovation, and applied research enables us to recognize outstanding achievements by both undergraduate and graduate students in engineering and computing. The Allen E. Paulson College of Engineering and Computing at Georgia Southern University sincerely appreciates their partnership and investment in supporting the next generation of engineers, technologists, and researchers. There are both graduate and undergraduate student divisions with awards presented as outlined below.

Crider Foods Manufacturing, Robotics and Automation Awards are presented for the top three posters in the combined categories of Robotics & Automation, AI/ML, Data Science, & Predictive Analytics, Manufacturing and Material Science.

Gulfstream Innovation Engineering and Flight Excellence Awards are presented for the top three posters in the combined categories of Robotics & Automation, Transportation & Automotive Technology, Manufacturing and Material Science, Power & Energy Systems and Simulation Modeling.

Gulfstream Completions Electrical and Computer Engineering Awards are presented for the top three posters Electrical and Computer Engineering.

Gulfstream Business Technology Computing Awards are presented for the top poster in Computer Science and to the top poster in Information Technology

Thomas and Hutton Civil Engineering and Construction Awards are presented for the top three posters in Civil Engineering and Construction.

Rolls-Royce Mechanical Engineering Awards are presented for the top three posters in Mechanical Engineering.

Allen E. Paulson Computing Awards are presented to the top three posters across all categories.

Allen E. Paulson Top Eagle Awards are presented to the top three posters across all categories.

Program Sponsors

Many thanks to our program sponsors for enabling the symposium to run smoothly and ensuring participants have a wonderful experience. This year's sponsorships support the use of the technology used in the judging process and for morning refreshments and afternoon refreshments.

Southern Nuclear - Morning Refreshment Sponsor

Port City Logistics - Afternoon Refreshment Sponsor

Poster Display Information

Poster 1-226

ID Number	Poster Name Department	Description	Session
AI/ML, Data Science, & Predictive Analytics (1-29)			
1	A Comparative Study of CNN and LSTM Models on American Sign Language Recognition Techniques **Computer Science**	This study compares CNN, LSTM, and hybrid CNN–LSTM models for American Sign Language recognition using the Kaggle ASL Alphabet dataset. CNNs capture spatial features, while LSTMs model temporal sequences. Performance is evaluated in terms of accuracy and parameter efficiency. The best-performing model achieved 87.2% accuracy, highlighting trade-offs between model complexity and performance. Findings inform the design of more effective and accessible ASL recognition systems.	UG
2	Building Unified Actor Timelines: A Temporal Framework for Analyzing APT Campaigns **Computer Science**	This project develops a unified temporal framework for analyzing Advanced Persistent Threat (APT) campaigns using MITRE ATT&CK and threat intelligence reports. Fragmented timestamps are normalized and aggregated into actor-level event streams, revealing temporal patterns such as bursts of activity and evolving tactics. The framework enables longitudinal analysis of adversary behavior and establishes a foundation for machine learning approaches to detecting and predicting emerging cyber threats.	UG
3	AI Navigator: Interactive Game Development for AI Literacy **Computer Science**	AI Navigator is a peer-designed, web-based educational game that promotes AI literacy and ethical awareness among undergraduates. Through interactive, scenario-based gameplay, students engage with real-world academic challenges involving AI tools. The project uses mixed-methods evaluation to assess engagement, decision-making, and learning outcomes. Results aim to demonstrate the effectiveness of game-based learning for improving responsible AI use and advancing scalable, student-centered digital literacy education.	UG

4	Window-Aware Clustering of Cryptocurrency Price Dynamics **Information Technology**	This study examines relationships among cryptocurrencies using time-series clustering across varying time windows. By analyzing how similarity patterns change with different temporal resolutions, the research reveals structural dynamics within the market. The focus is on identifying stable and shifting relationships among assets rather than forecasting prices. Findings contribute to a deeper understanding of cryptocurrency market behavior and interdependencies.	UG
5	A Quantitative Analysis of Computer Vision Systems' Reliability in Detecting Human Emotions **Computer Science**	This study evaluates the reliability of a facial landmark–based emotion detection system using OpenCV and MediaPipe. Geometric features such as mouth curvature and eyebrow position are used for classification. Results show strong performance for exaggerated expressions but reduced accuracy for subtle emotions. The findings highlight limitations of rule-based approaches and underscore challenges in replicating human emotional perception in computer vision systems.	UG
6	Project Akira: Data Mining and Merging, and training a Custom-Built GPT to produce Therapist-Like, Ethically Coherent Responses **Computer Science**	This project enhances a custom-built GPT model (AkiraGPT) by integrating merged mental-health dialogue datasets and retraining with optimized parameters. The goal is to improve ethical coherence, contextual relevance, and response completeness. The study examines how dataset composition and training adjustments affect model behavior, providing insights into building more reliable and responsible conversational AI systems.	UG
7	Adaptive image-based classification of concrete rebars and defects in ground-penetrating radargrams **Electrical & Computer Engineering**	ConcreteNet is a convolutional neural network framework for analyzing ground-penetrating radar (GPR) data to detect rebars and structural defects in concrete. Trained on curated and augmented datasets, the model demonstrates reliable identification of subsurface features. Validation with real-world inspection data highlights its potential for automated infrastructure assessment. The project also provides an open dataset to support benchmarking and reproducibility in nondestructive evaluation research.	UG

8	Computational Analysis of Impressionist Visual Structure **Computer Science**	This study applies convolutional neural networks to classify Impressionist artwork based on texture, color, and structural features. By identifying distinguishing-visual characteristics, the model demonstrates the feasibility of automated art style classification. The work contributes to computational art history and supports scalable analysis of digitized art collections.	UG
10	Patterns in AI-Driven Misinformation: Comparing Chat-GPT and Gemini **Information Technology**	This research compares misinformation patterns generated by large language models, specifically ChatGPT and Gemini. It analyzes differences in accuracy, language sophistication, and hallucination tendencies. The study identifies model-specific vulnerabilities and contributes insights for improving misinformation detection, governance strategies, and digital literacy efforts in AI-driven information environments.	GRAD
11	Machine Learning Models for Early Prediction of Cardiovascular Risk in Patients with Type 2 Diabetes **Information Technology**	This study develops machine learning models to predict cardiovascular risk in patients with Type 2 Diabetes using a dataset of 70,000 individuals. Logistic Regression, Random Forest, and SVM models are evaluated, with Logistic Regression achieving the highest performance (AUC ≈ 0.80). Key predictors include blood pressure, BMI, and lifestyle factors. Results demonstrate the potential of ML to support early risk stratification and clinical decision-making.	GRAD
12	Negative Impact of Social Media on Mental Health **Information Technology**	This study analyzes the relationship between social media use and mental health outcomes using machine learning models. Multi-output classification approaches outperform single-task models in predicting depression, distraction, and sleep issues. Results highlight the interconnected nature of mental health effects and demonstrate the value of modeling multiple outcomes simultaneously for improved predictive accuracy.	GRAD

13	Detection of COPD-like Breathing Patterns from Non-Invasive Pressure and Flow Data Using Machine and Deep Learning. **Information Technology**	This research uses machine learning and deep learning models to detect COPD-like breathing patterns from non-invasive pressure and flow signals. Models achieve up to 98% accuracy, with interpretability techniques identifying key physiological features. The framework supports scalable, automated respiratory screening and offers a non-invasive alternative to traditional diagnostic methods.	GRAD
14	TRACE: Transparent Recognition of Affective Cues via Expression **Computer Science**	TRACE investigates microexpression recognition using spatiotemporal modeling and facial action units. The research focuses on capturing subtle, involuntary expressions that reveal true emotional states. By emphasizing interpretability and biologically meaningful features, the study advances data-efficient and transparent approaches to affective computing and human-centered AI.	GRAD
15	An Automated and Explainable AI Pipeline for Molecular Descriptor-Based QSAR Modeling **Information Technology**	This study presents an automated, explainable machine learning pipeline for QSAR-based drug activity prediction using molecular descriptors. Multiple models are benchmarked, with a blended ensemble achieving strong performance (AUC 0.90). Interpretability tools and applicability domain analysis ensure reliability. The pipeline advances reproducible and transparent approaches in predictive toxicology and cheminformatics.	GRAD
16	A Study on How Data Quality Affects Predictive Analytics: Insights from Healthcare and Retail **Information Technology**	This research examines how data quality issues—such as missing values, noise, and inconsistencies—affect predictive analytics in healthcare and retail. Experimental results show that preprocessing techniques significantly improve model accuracy and reliability. The study bridges theoretical data quality concepts with practical applications, emphasizing the importance of clean data for effective decision-making.	GRAD
17	Security-Oriented Voice Authentication Using Machine Learning **Information Technology**	This study develops a machine learning-based voice authentication system using features such as MFCCs and spectral characteristics. Models including SVM, Random Forest, and CNN are evaluated using standard performance metrics. Results demonstrate strong accuracy and reliability, supporting the use of voice biometrics in secure authentication systems.	GRAD

18	A Digital Twin Framework for Cyber-Physical System Detection in Water Treatment Infrastructure **Electrical & Computer Engineering**	This paper presents a digital twin framework combining discrete-event simulation and LSTM models to detect cyber-physical attacks in water treatment systems. The approach achieves over 92% accuracy in identifying and classifying simulated attacks. Results demonstrate the potential of digital twins for enhancing infrastructure security and operational monitoring.	GRAD
19	Probabilistic Spatio-Temporal Neural Modeling for Flight Delay Prediction **Computer Science**	This study proposes a hybrid machine learning framework for predicting flight delays using operational, network, and weather data. A two-stage approach estimates delay probability and magnitude using classification and neural regression. The model captures temporal dynamics and complex feature interactions, improving prediction accuracy and supporting airline operations.	GRAD
20	Detecting Suspicious Blockchain Transactions Through Decentralized Identifier-Framework-Inspired Data Augmentation **Information Technology**	This research enhances blockchain fraud detection by augmenting transaction data with synthetic identity features inspired by decentralized identifiers. Machine learning models, including anomaly detection techniques, are evaluated using standard metrics. Results demonstrate improved detection performance, highlighting the value of enriched feature engineering.	GRAD
21	Evaluating Deep Learning Architectures for Actin Filament Segmentation under Varying Noise Conditions in Simulated Cryo-ET Tomograms **Computer Science**	This study evaluates deep learning architectures for segmenting actin filaments in simulated cryo-ET data under varying noise conditions. Models are assessed using Dice and IoU metrics. Results show no single architecture consistently outperforms others, emphasizing the importance of adapting models to data characteristics.	GRAD

22	CA-IDD (Cross-Attention Guided Identity-Conditional Diffusion): A Robust and Controllable Framework for Realistic Face Swapping **Information Technology**	This paper introduces CA-IDD, a diffusion-based face-swapping framework using cross-attention to improve identity consistency and realism. The model achieves strong performance (FID 11.73) and demonstrates robust generalization across poses and expressions, outperforming GAN-based approaches.	GRAD
23	Enhancing Fraud Detection Using Machine Learning **Computer Science**	This study develops a stacked ensemble model combining Logistic Regression and Random Forest for credit card fraud detection. The approach improves performance under class imbalance while maintaining interpretability. Results show enhanced precision and recall, offering a practical solution for real-world financial fraud detection.	GRAD
Robotics & Automation (30-59)			
30	Development of a Pilot AI for Autonomous UAV Inspection in Industry 5.0 using Visual SLAM and Behavior Trees **Mechanical Engineering**	This research develops an autonomous UAV inspection system integrating Visual SLAM and Behavior Tree–based AI for navigation in GPS-denied industrial environments. A hardware platform processes real-time sensor data to enable localization and adaptive flight control. The system aims to autonomously perform facility inspections, reducing downtime and human intervention while validating real-world deployment of AI-driven UAV oversight.	UG
31	Low-Cost Centimeter-Level Robot Localization Using Existing Infrastructure and Technologies **Information Technology**	This study proposes a low-cost robot localization method using fixed infrastructure cameras instead of onboard sensors. By applying OpenCV-based geometric transformations and landmark calibration, the system achieves 1–3 cm accuracy under controlled conditions. The approach reduces cost and complexity, supports multi-robot coordination, and demonstrates how existing camera networks can enable precise, scalable localization in GPS-denied environments.	UG

32	Putting Robots Where Humans Shouldn't Be: AI-Enabled Humanoid Robotics for Hazard Detection **Electrical & Computer Engineering**	This project develops an AI-enabled humanoid robot for hazard detection in dangerous environments such as construction sites and industrial facilities. The system integrates multisensor inputs and onboard AI to detect hazards, perform inspections, and communicate risks. Its human-like form enables navigation in built environments, improving safety, situational awareness, and operational efficiency while reducing human exposure to hazardous conditions.	UG
33	Robust Curb Detection on Unmarked Roads Using LiDAR Using a Rule-Based Algorithm **Mechanical Engineering**	This research presents a LiDAR-based curb detection framework for unstructured and degraded road environments. The system identifies curb boundaries using geometric features such as height variation and surface normals, without relying on maps or visual markings. A robust processing pipeline handles noise and irregular terrain, enabling reliable detection in challenging conditions and improving autonomous vehicle navigation in poorly defined road environments.	UG
34	Training AI-Based Detection Models to Analyze Defects in Canned Goods **Mechanical Engineering**	This study investigates defect detection in canned goods by combining RGB imagery with depth sensing. A multimodal perception pipeline identifies surface deformations such as dents and bulges more accurately than RGB-only methods. Preliminary results suggest improved detection performance, with further work focusing on data collection across varying conditions to enhance model robustness and sensitivity to subtle defects.	UG
35	Sensor Fusion-Based Obstacle-Aware Trajectory Replanning for Robotic Arms **Electrical & Computer Engineering**	This project develops a robotic arm system that uses sensor fusion to detect obstacles and dynamically adjust its trajectory in real time. Integrating infrared and optical sensors with kinematic modeling, the system enables proactive collision avoidance without interrupting operation. Simulations validate performance, demonstrating improved safety and flexibility for industrial and collaborative robotic applications.	UG

36	Vivarium Automated Control System **Electrical & Computer Engineering**	This project designs an automated system to regulate temperature and humidity in reptile and amphibian habitats. Using an Arduino-based platform with sensors, relays, and wireless control, the system maintains stable environmental conditions. The solution improves reliability, reduces manual monitoring, and demonstrates the potential for affordable smart habitat systems in animal care and environmental monitoring applications.	UG
37	EnCompass: Electronic Component Analysis and Screening System **Electrical & Computer Engineering**	This project develops a portable device for automatic identification and testing of electronic components. Using a microcontroller-based system, it applies test signals to classify components and measure values, displaying results on an LCD. The system simplifies diagnostics, improves accessibility, and provides a cost-effective alternative to traditional testing equipment for educational and prototyping environments.	UG
38	Digital Keyboard Implemented by Integrated Gloves **Electrical & Computer Engineering**	This project develops a wearable keyboard system using sensor-integrated gloves. Bend and compass sensors detect finger movements, while an Arduino and Bluetooth module transmit inputs. The system enables typing on flat surfaces with improved ergonomics and flexibility. The prototype demonstrates feasibility, with ongoing work focused on accuracy, responsiveness, and expanded functionality.	UG
39	Application of Collaborative Robotics and Artificial Intelligence for Adaptive Path Planning in Welding Inspection **Mechanical Engineering**	This study integrates collaborative robotics, ultrasonic testing, and AI to enable adaptive welding inspection. The system dynamically adjusts scanning paths based on detected flaws, improving efficiency and detection accuracy compared to predefined methods. MATLAB-based analysis supports real-time flaw identification, enhancing reliability, repeatability, and automation in nondestructive testing applications.	UG

40	Will Robots Take Our Jobs? Automation, Employment, and Workforce Change **Computer Science**	This research examines the impact of automation on employment and workforce dynamics. Using labor data and case studies, it analyzes how robotics influences job availability, skill requirements, and productivity. Findings highlight both risks and opportunities, emphasizing the importance of education and reskilling to support workforce adaptation in an increasingly automated economy.	UG
41	Deep Learning–Powered 3D Vision for In-Line Robotic Inspection in Manufacturing **Manufacturing Engineering**	This study integrates stereo vision and deep learning to generate 3D reconstructions for robotic inspection. Using FoundationStereo and Open3D, the system produces dense point clouds from multi-view images. The approach supports automated quality inspection and metrology, with evaluation metrics assessing reconstruction accuracy for complex surfaces in manufacturing environments.	UG
50	An Autonomous Mobile Robotic System for Crack Detection and 3D Profiling in Infrastructure Maintenance **Civil Engineering & Construction**	This research presents an autonomous robotic system for infrastructure inspection combining SLAM navigation, deep learning-based crack detection, and 3D profiling. Using U-Net and GAN models, the system achieves accurate segmentation and geometric analysis of cracks. The integrated platform enables automated scan–assess–repair workflows, improving safety and efficiency in infrastructure maintenance.	GRAD
51	Digital Reconstruction of Structural Damage Using the Spot Robotic Platform **Civil Engineering & Construction**	This study evaluates the use of the Spot robotic platform for automated documentation and reconstruction of structural damage. Computer vision techniques generate orthomosaics and 2.5D models from repeatable image capture. Results demonstrate the potential of robotic systems for consistent, quantitative forensic analysis in engineering applications.	GRAD
52	Improved Hopping Control on Slopes for Small Robots Using Spring Mass Modeling **Electrical & Computer Engineering**	This study improves hopping robot stability on slopes using a spring-mass model. By adjusting touchdown angle and applying corrective torque, the method compensates for slope-induced rotation. Simulations show improved landing stability with minimal computational requirements, enabling reliable locomotion on uneven terrain for low-cost robotic platforms.	GRAD

Cybersecurity & Data Protection and Networking & Communication Systems (60-79)			
60	Multi-environment Wi-Fi CSI-based Human Fall Detection with CNN **Electrical & Computer Engineering**	This project develops a privacy-preserving fall detection system using Wi-Fi channel state information (CSI). Signal processing techniques and a CNN model analyze motion patterns without cameras. The system targets over 80% detection accuracy and offers a non-intrusive solution for elderly care environments, balancing effectiveness with user privacy while enabling real-time fall monitoring.	UG
61	Search and Rescue System using Channel State Information (CSI) Packets **Electrical & Computer Engineering**	This project designs a CSI-based system for detecting human presence in obstructed environments such as collapsed structures. Using ESP32 nodes and RF sensing, the system captures signal variations to identify motion and displays results as heatmaps. Preliminary results confirm feasibility, with ongoing work focused on improving localization accuracy and extending to multi-dimensional detection scenarios.	UG
62	Hybrid Sensor and Vision-Based Smart Parking Availability System **Electrical & Computer Engineering**	This study presents a hybrid parking detection system combining ultrasonic sensors and camera-based monitoring. A cloud-connected mobile application provides real-time parking availability to users. The prototype demonstrates reliable occupancy detection and communication performance, offering a scalable solution to reduce congestion and improve efficiency in high-density environments such as campuses and urban areas.	UG
63	Web-Based Windows Hardening Platform with CIS Compliance **Information Technology**	This project develops a web-based platform to automate Windows 11 system hardening using CIS Level 1 benchmarks. Leveraging Wazuh for security configuration assessment and Flask for backend integration, the system provides real-time compliance monitoring through an interactive dashboard, improving efficiency, visibility, and consistency in cybersecurity configuration management.	UG

70	Detecting Deepfakes through Visual Property Analysis and Classical Machine Learning Models **Information Technology**	This study proposes an interpretable deepfake detection system using handcrafted visual features and classical machine learning models. Features include texture, color, and frequency-domain artifacts. Models such as Random Forest and XGBoost achieve strong accuracy with low computational cost. Results highlight the effectiveness of lightweight, explainable approaches for practical and resource-constrained forensic applications.	GRAD
71	WebAssembly-Based AI-Driven Framework for Real-Time Preemptive Detection of Zero-Day Browser Exploits **Information Technology**	This research introduces a browser-integrated framework for detecting zero-day exploits using WebAssembly telemetry and unsupervised learning. An Isolation Forest identifies anomalous behavior, while a large language model provides contextual explanations. The system enables real-time, preemptive detection of malicious activity, enhancing browser security without relying on known attack signatures.	GRAD
72	Performance Evaluation of a Load-Balanced Virtual Network Architecture Using JMeter in a Multi-Tenant Environment **Information Technology**	This study evaluates a virtual load-balanced network architecture using Apache JMeter. Compared to a single-server setup, the load-balanced system significantly reduces latency, improves response time consistency, and eliminates errors under high traffic. Results demonstrate the effectiveness of load balancing in enhancing scalability, reliability, and performance in multi-tenant virtual environments.	GRAD
73	Leveraging Machine Learning for Phishing Website Detection Using URL and Domain-Level Features **Information Technology**	This study evaluates machine learning models for phishing detection using URL and domain-level features. Random Forest achieves the highest performance, though results decline under temporal zero-day conditions. Findings emphasize the importance of evaluating generalization and demonstrate the effectiveness of lightweight features for scalable phishing detection systems.	GRAD

74	Closed-Loop Cyber Defense Using Agentic AI with Continuous Feedback and Learning **Information Technology**	This research presents a closed-loop cyber defense system using Agentic AI for continuous monitoring, decision-making, and response. The system classifies attack stages and adapts mitigation strategies through feedback. Evaluation shows high effectiveness in detecting and responding to threats, demonstrating improved adaptability and consistency in modern cybersecurity environments.	GRAD
75	A Unified Cyber-Resilient Control Framework for Shunt Active Power Filters in Grid-Tied Photovoltaic Systems **Electrical & Computer Engineering**	This study proposes a cyber-resilient control framework for SAPFs in grid-tied photovoltaic systems. The approach detects and mitigates cyberattacks affecting control stability, preserving harmonic compensation and voltage regulation. Results show improved resilience under adversarial conditions, ensuring stable and reliable power system operation.	GRAD
76	Performance Analysis of LTE-V2V Mode 4 Communication Using SPS Resource Allocation **Electrical & Computer Engineering**	This research evaluates LTE V2V Mode 4 communication performance using semi-persistent scheduling. The study analyzes system behavior under realistic conditions and assesses feasibility for autonomous vehicle communication without infrastructure. Results highlight performance strengths and limitations, contributing to improved design of vehicle-to-vehicle communication systems for safety-critical applications.	GRAD
Structural & Civil Engineering (80-99)			
80	Building Simultaneous Resilient and Sustainable Roadway Infrastructure **Civil Engineering & Construction**	This study proposes a unified Quality Prioritization Index (QPI) integrating resilience and sustainability for roadway systems using FAHP. Key resilience factors were identified and weighted, then combined with sustainability metrics. A case study across Gulf Coast states demonstrates the framework's effectiveness in prioritizing interventions, supporting data-driven decision-making for infrastructure planning under climate and aging challenges.	UG

81	Student Steel Bridge Competition (SSBC) **Civil Engineering & Construction**	This project involves the design and construction of a steel pedestrian bridge for the SSBC competition. Students apply structural analysis, fabrication, and construction planning while meeting performance, cost, and aesthetic criteria. The project provides hands-on experience in real-world engineering design, emphasizing efficiency, safety, and sustainability.	UG
82	Investigation of Internal Structural Integrity of Aircraft Wings **Mechanical Engineering**	This research investigates factors affecting aircraft wing structural integrity, including fatigue, vibration, impact loads, and design configurations. Focusing on ribs and spars, the study examines degradation mechanisms using experimental and analytical approaches. The goal is to improve durability, safety, and performance of aircraft structures over their lifecycle.	UG
83	Compatibility of SBS Binder with PET and GTR Asphalt Mixtures **Civil Engineering & Construction**	This study evaluates the compatibility of SBS-modified binders with PET and GTR additives in asphalt mixtures. Testing includes compaction, rutting, and cracking performance. Results suggest improved compactibility and durability, supporting sustainable pavement design through the use of recycled materials.	UG
84	Timber Strong Design Build 2026 **Civil Engineering & Construction**	This project involves designing and constructing a sustainable wood-framed structure for competition. Students analyze loads, optimize structural performance, and incorporate sustainability principles. The project provides comprehensive experience in design, construction, and environmental considerations in modern timber engineering.	UG
85	2025-2026 ASCE Concrete Canoe Competition **Civil Engineering & Construction**	This project involves designing and constructing a lightweight concrete canoe for competition. The canoe meets structural, aesthetic, and performance criteria while supporting racing conditions. The project demonstrates innovation in materials engineering and project management, showcasing the feasibility of unconventional concrete applications.	UG

86	Design and Construction of a Small-Scale MSE Wall **Civil Engineering & Construction**	This project designs and builds a small-scale mechanically stabilized earth wall under specified loads and constraints. Using scaled reinforcement and geotechnical principles, the structure demonstrates stability under surcharge and obstruction conditions, providing hands-on experience in earth-retaining system design.	UG
90	Flood Risk-Informed Life Cycle Analysis of Water Distribution Systems **Civil Engineering & Construction**	This study develops a risk-informed framework for evaluating life cycle costs of water distribution systems under flooding conditions. By integrating likelihood and consequence of failure, the model identifies high-risk assets and supports optimized maintenance planning, improving resilience of critical infrastructure.	GRAD
91	Evaluation of High-Temperature Performance of Asphalt Mixtures **Civil Engineering & Construction**	This research evaluates IDEAL-RT testing as a predictor of rutting resistance in asphalt mixtures. Strong correlations with advanced performance metrics suggest the method's effectiveness as a practical screening tool for mixture design and quality control.	GRAD
92	Multiple Short Breaks vs. One Long Break in Worker Fatigue **Civil Engineering & Construction**	This study analyzes the effects of rest intervals on worker fatigue using wearable sensor data and machine learning. Results show multiple short breaks significantly improve physiological recovery compared to one long break, supporting improved safety and performance in labor-intensive environments.	GRAD
93	Pile Setup Effects on Driven Piles in Southeastern Coastal States **Civil Engineering & Construction**	This study analyzes pile setup behavior using load test data from southeastern U.S. sites. Results highlight variability across soil conditions and emphasize the importance of time-dependent resistance in design. Findings support improved reliability in foundation engineering practices.	GRAD
94	Localized LRFD Resistance Factor Calibration for Driven Piles **Civil Engineering & Construction**	This research calibrates LRFD resistance factors for driven piles in Georgia using dynamic and static test data. Reliability analyses improve design accuracy and provide region-specific recommendations for foundation engineering under varying soil conditions.	GRAD

95	Digital Twin–Enabled Evaluation of Over-Height Vehicle Detection Systems **Civil Engineering & Construction**	This study evaluates a bridge over-height vehicle detection system using a digital twin framework. Integrating sensor data and virtual modeling enables real-time monitoring and performance analysis, improving system reliability and supporting proactive infrastructure maintenance.	GRAD
96	Remote Sensing-Based Displacement Monitoring of MSE Retaining Walls **Civil Engineering & Construction**	This research compares LiDAR and photogrammetry methods for monitoring MSE wall displacement. Results show TLS provides millimeter-level accuracy, while photogrammetry has higher detection limits. The framework supports reliable, data-driven condition assessment for infrastructure monitoring.	GRAD
Transportation & Automotive Technology (100-119)			
100	Measuring Real Time Combustion Chamber Pressure to Increase Fuel Efficiency and Decrease Green House Gas Emissions **Mechanical Engineering**	This project measures in-cylinder combustion pressure using piezoelectric sensors to evaluate engine performance and emissions. Metrics such as apparent heat release rate and indicated mean effective pressure are derived to diagnose inefficiencies, friction losses, and abnormal combustion. Results support improved engine tuning strategies to enhance fuel efficiency and reduce greenhouse gas emissions in internal combustion engines.	UG
101	Improving Diesel Engine Fuel Consumption and Emissions Through Flow Meter Calibration **Mechanical Engineering**	This study improves diesel engine efficiency by calibrating a piston flow meter to measure fuel consumption and calculate brake specific fuel consumption (BSFC). By accurately quantifying fuel usage, the system enables optimization of engine performance, reducing emissions and improving energy efficiency. The work supports sustainable engine operation through better instrumentation and calibration.	UG

102	Supercharger Integration into Diesel Engine Design Efficiency **Mechanical Engineering**	This research investigates supercharger integration to improve diesel engine efficiency and reduce emissions. By increasing intake air, combustion becomes more complete, enhancing energy extraction and reducing fuel consumption. The study evaluates supercharger configurations on a single-cylinder test platform to expand operating range and improve performance for cleaner combustion systems.	UG
103	Implementation of Steer-by-Wire on a Hybrid Vehicle **Mechanical Engineering**	This project designs and implements a steer-by-wire system on a hybrid vehicle, replacing mechanical steering with electronic control. Using sensors, actuators, and embedded systems, the design enables adjustable steering response and reduced system weight. The prototype demonstrates reliable control and highlights advantages of modern electronic steering architectures.	UG
104	Supersonic Combustion **Mechanical Engineering**	This study examines supersonic combustion processes relevant to scramjet propulsion. It highlights challenges such as rapid mixing, flame stabilization, and shock interactions in high-speed flows. Experimental and computational methods are reviewed, along with design strategies for improving combustion efficiency in hypersonic propulsion systems.	UG
105	Aerodynamic Studies of Blended Wing Body **Mechanical Engineering**	This research evaluates the aerodynamic performance of blended wing body (BWB) aircraft configurations. By integrating wing and fuselage, BWB designs reduce drag and improve fuel efficiency. Results indicate significant fuel savings compared to conventional aircraft, supporting the potential for sustainable, large-capacity transport designs.	UG
110	Information-Driven Calibration of Vehicle Simulations for Precise Control Response **Mechanical Engineering**	This study develops a calibration framework for aligning vehicle simulations with real-world behavior. Using experimental data and LSTM models, the system predicts vehicle response and adjusts simulation parameters. The approach improves accuracy of control systems developed in simulation, enhancing reliability for autonomous and advanced driver-assistance systems.	GRAD

111	Accuracy Comparison of LiDAR and Photogrammetry for 3D Mapping **Mechanical Engineering**	This research compares LiDAR and photogrammetry for 3D mapping of bridge infrastructure. Using GPS-based reference data, model accuracy and drift are evaluated across long distances. Results provide guidance for integrating sensing technologies in digital twin applications and infrastructure monitoring.	GRAD
Environmental & Sustainability (120-139)			
120	ENVISION-AI – ENVirOnmental ModellIng, predictIon & forecaSting with AI **Computer Science**	This project uses AI to model and forecast PM2.5 air pollution from traffic, wildfire, and industrial sources. By integrating large-scale environmental sensor data, the system captures complex spatiotemporal patterns, supports uncertainty-aware prediction, and enables anomaly detection and early warnings. The framework is intended to improve air-quality monitoring, inform policy, and strengthen public-health protection and community resilience.	UG
121	Assessing Microplastic Contamination and Water Quality Across Natural and Human-Impacted Freshwater Systems in Costa Rica **Civil Engineering & Construction**	This study investigates microplastic contamination in Costa Rican freshwater systems spanning pristine, peri-urban, and agricultural watersheds. It combines water-quality measurements, trace-element analysis, microbial profiling, micro-Raman polymer identification, and community surveys. By linking environmental data with public perceptions, the project establishes a baseline for tropical freshwater pollution and supports integrated strategies for monitoring, management, and public engagement.	UG
130	Mapping and Prioritizing Barriers to Successful Rural Recovery Affected by Hurricanes **Civil Engineering & Construction**	This research identifies and prioritizes barriers to post-hurricane recovery in rural communities using literature review, expert validation, and social network analysis. Fifty-two barriers were classified and ranked by centrality and interconnectedness, producing a seven-level framework for intervention. The results help policymakers target high-leverage barriers, improve recovery efficiency, and promote more equitable restoration of vulnerable rural areas.	GRAD

131	Interrelationships among Barriers to Timely Recovery after Hurricane: Structural Equation Modelling Approach **Civil Engineering & Construction**	This study examines how interconnected recovery barriers delay post-hurricane recovery in rural communities. Using literature review, expert input, and structural equation modeling, it analyzes 52 barriers across eight categories and their effects on recovery timeliness. Results show that economic and infrastructure barriers have the strongest direct impacts, while governance and management are highly correlated, informing more effective resource allocation and recovery planning.	GRAD
132	Charting Synergistic Pathways for Increased Water Efficiency in Coastal Georgia **Civil Engineering & Construction**	This project evaluates water-use efficiency, conservation potential, and reuse opportunities across municipal, industrial, agricultural, and recreational sectors in Coastal Georgia. Using surveys, focus groups, and quantitative analysis, it identifies current use patterns, barriers, and promising interventions. Key outputs include a regional water-use dataset, ranked efficiency projects, and implementation roadmaps to support sustainable growth, policy development, and long-term water security.	GRAD
133	A Universal AI-IoT Integrated System for Adaptive Water Quality Assessment Using Remote Sensing and Multi-Context WQI Modeling **Information Technology**	This research proposes an AI-IoT framework for adaptive water-quality assessment across diverse aquatic environments. The system combines real-time sensors, remote sensing, and hybrid machine-learning models to generate context-specific Water Quality Index outputs for uses such as drinking, irrigation, and aquaculture. With explainable AI, hazard forecasting, and treatment recommendations, the framework supports scalable, intelligent water monitoring and decision-making.	GRAD
134	Mapping Septic System Vulnerability and Microbial Contamination Pathways in Coastal Georgia (Sapelo Island) **Civil Engineering & Construction**	This study integrates GIS analysis and field sampling to assess septic-system vulnerability and microbial contamination pathways on Sapelo Island. Spatial mapping identifies overlap among septic systems, wetlands, and flood-prone zones, while soil and water testing measures microbial and physicochemical indicators. Results show strong links between landscape position, hydrologic connectivity, and contamination risk, supporting improved septic management and coastal water-quality protection.	GRAD

136	Nationwide Sustainability Assessment of Granular Activated Carbon and Ion Exchange for PFAS Safe Drinking Water in U.S. Utilities **Civil Engineering & Construction**	This study evaluates granular activated carbon and ion exchange for PFAS removal in U.S. drinking water systems using integrated techno-economic and life-cycle assessment. Monte Carlo and sensitivity analyses show utility size and PFAS chain length strongly influence cost and environmental impacts. Results highlight tradeoffs between treatment options and reveal disproportionate burdens on small and rural systems, informing more equitable compliance strategies.	GRAD
Biomedical, Health, & Computer Interface (140-169)			
140	OncoTrack - Oncology Malignant Cell Tracking & Performance Validation **Computer Science**	This project develops a real-time particle tracking system inspired by cancer cell migration. Using video data, the system tracks particle trajectories, computes velocity and acceleration fields, and generates statistical metrics. The approach enables dynamic analysis of movement patterns, supporting research in biological systems and providing a flexible tool for tracking and analyzing motion in complex environments.	UG
141	DeepMedSeg- Deep Learning for Medical Image Segmentation **Computer Science**	This study develops a deep learning framework using U-Net architecture to automate medical image segmentation. The system aims to accurately identify anatomical and pathological regions in imaging data, reducing reliance on manual annotation. Results target improved precision, efficiency, and consistency, supporting clinical workflows such as diagnosis, treatment planning, and disease monitoring.	UG
142	Recipe for Code: Immersive Java Gaming **Computer Science**	This project introduces a 2D cooking-themed game to teach Java programming concepts through interactive gameplay. Coding principles such as loops and algorithms are embedded in real-world scenarios, improving engagement and retention. The system supports inclusive learning, offering a hands-on, narrative-driven approach to computer science education for beginners.	UG
143	Improving the Modal Characteristics of 3D Bioprinter By Passive and Active Vibration Control	This research addresses vibration-induced defects in 3D bioprinting by applying mechanical isolation and signal processing techniques. The goal is to reduce print bed resonance, improving print accuracy and enabling clearer frequency analysis of bio-printed	UG

	Mechanical Engineering	structures. Results support more reliable bioprinting processes for tissue engineering applications.	
144	NFC Access for Medical Questionnaires **Information Technology**	This project develops an NFC-based system for accessing medical questionnaires using tokenized identifiers linked to a centralized database. The solution reduces redundant data entry, improves workflow efficiency, and enhances accuracy. A full-stack prototype demonstrates seamless integration of hardware and software for modern healthcare intake systems.	UG
145	3D Transient Magneto-Fluid-Structure Interaction Modeling of Magnetorheological Micropumps with Validation **Mechanical Engineering**	This research develops a 3D simulation model for magnetorheological micropumps used in insulin delivery systems. The model predicts fluid flow and performance under wearable constraints, supporting design optimization. Results demonstrate feasibility for compact, low-power drug delivery devices and broader biomedical applications.	UG
145	Microplastic Particles Emitted from 3D Printing and Their Mitigation **Civil Engineering & Construction**	This study evaluates microplastic emissions from 3D printers using optical particle sensors under different conditions. Results assess exposure risks and test mitigation strategies such as air purification. Findings provide insight into environmental and health impacts of additive manufacturing.	UG
147	Process-Property-Geometry Tuning of Magnetorheological Elastomers for Insulin-Delivery Micropumps **Mechanical Engineering**	This project investigates optimization of magnetorheological elastomers for micropump applications. By varying material composition and fabrication methods, the study improves mechanical and magnetic performance. Results support enhanced efficiency and durability for insulin delivery systems.	UG
148	Development of a Cost Effective and Open-Source Modular Bio-3D Printer	This project develops a low-cost, modular bio-3D printer using modified conventional systems. The design emphasizes accessibility, flexibility, and open-source architecture, enabling broader adoption in research and education while supporting innovation in tissue engineering.	UG

	Mechanical Engineering		
149	A 3D Scan Based Patient Specific Device for At Home Volumetric Monitoring of Knee Swelling **Mechanical Engineering**	This study proposes a patient-specific device using 3D scanning and volumetric measurement to monitor knee swelling in rheumatoid arthritis. The approach enables at-home tracking, improving early intervention and supporting continuous clinical assessment.	UG
150	Bioimpedance Spectroscopy Device Improvements Through Different Circle Fit Methods **Mechanical Engineering**	This research improves bioimpedance spectroscopy accuracy by evaluating alternative circle-fitting methods. The study enhances resistance estimation reliability, supporting better body composition analysis and clinical applications while improving computational efficiency.	UG
151	Low-Cost Velocity Estimation Using IMU Sensor Fusion and Computer Vision for Gymnastics **Mechanical Engineering**	This study develops a low-cost system combining IMU sensor fusion and computer vision to improve velocity estimation in athletic motion. The approach reduces drift and noise, enabling accurate performance analysis for gymnastics and other applications.	UG
152	Portable MAX-30009 Based Bio-Impedance Measurement Device **Mechanical Engineering**	This project develops a portable bio-impedance device using MAX-30009 hardware with improved accuracy through backend calibration. The system provides reliable multi-frequency measurements for health monitoring applications.	UG
153	Accessible Card Game Interaction System **Electrical & Computer Engineering**	This project designs an automated card game system to support individuals with cognitive disabilities. The system shuffles and deals cards using microcontroller-based controls, improving accessibility, engagement, and therapeutic interaction.	UG

160	Bio-sense: A Multimodal Sensor-Based Prototype for Health Abnormality Detection **Information Technology**	Bio-Sense integrates multiple sensors to monitor physiological signals and detect health abnormalities in real time. Using machine learning, the system analyzes ECG, SpO2, and other signals, providing a scalable, noninvasive health monitoring solution.	GRAD
161	Early ICU Physiological Subtypes from Time-Series Data and Patient Prognosis **Information Technology**	This study identifies ICU patient subtypes using time-series physiological data and machine learning. Results show distinct patterns linked to mortality risk, improving prognostic modeling and supporting personalized clinical decision-making.	GRAD
162	Emerging AI-based Tools and Techniques for ADHD Detection **Computer Science**	This review examines AI and eye-tracking methods for ADHD detection in college students. Results highlight potential for objective, scalable screening, while identifying challenges such as data variability and lack of standardization.	GRAD
163	Co-axial electrospraying of injectable multi-cancer drugs nano-capsules **Mechanical Engineering**	This study develops polymer-based nanocapsules using coaxial electrospraying for targeted cancer therapy. Results demonstrate improved drug delivery, sustained release, and enhanced treatment outcomes for aggressive breast cancers.	GRAD
164	Quantum-Enhanced Noninvasive Approach for Continuous Glucose Monitoring **Information Technology**	This research proposes a quantum-enhanced, noninvasive glucose monitoring system using optical sensing and machine learning. The approach improves sensitivity and enables continuous, needle-free glucose tracking, supporting patient-friendly diabetes management.	GRAD
Manufacturing & Material Science (170-189)			

170	Environmentally Friendly LED Light-Curable and Self-Healable Materials **Mechanical Engineering**	This study develops a self-healing coating that cures under visible LED light and repairs damage via photothermal effects. Using eco-friendly monomers and nanofillers, the material rapidly cures and heals scratches within minutes. The approach reduces energy use and avoids hazardous triggers, offering a sustainable solution for protective coatings in demanding environments.	UG
171	Adaptive Contact Force Control for Robotic PAUT Inspection of Welded Steel **Mechanical Engineering**	This research develops a robotic system with machine vision and force feedback to improve ultrasonic inspection of welded steel. Adaptive control ensures consistent probe contact, enhancing signal quality and defect detection accuracy. Results demonstrate improved reliability compared to manual inspections, supporting scalable automated quality assurance.	UG
172	Application of Machine Vision for In-situ Monitoring in Additive Manufacturing **Mechanical Engineering**	This study applies machine vision and YOLO-based models to detect defects during additive manufacturing. By overcoming limitations of fixed camera angles, the system improves identification of surface flaws. Results demonstrate enhanced accuracy in defect classification, supporting real-time quality control and broader industrial adoption of AM.	UG
173	Shrinkage Prediction of BCC Metal Lattices Fabricated by FFF Additive Manufacturing **Manufacturing Engineering**	This research investigates shrinkage behavior in FFF-fabricated metal lattice structures. Using design of experiments and microscopy, the study identifies key parameters affecting dimensional accuracy. A predictive model is developed to guide design and improve reliability of low-cost metal additive manufacturing.	UG
174	Topology Optimization of 3D Printed Composite Pi-Joints **Mechanical Engineering**	This study uses topology optimization to design lightweight composite pi-joints under multi-axial loading. By optimizing load paths, the design reduces material usage while maintaining strength and stiffness. Results demonstrate efficient, cost-effective structures for aerospace and mechanical applications.	UG

180	Instant-Cure Methacrylate-Based Matrices for Fiber-Filled Composites **Mechanical Engineering**	This project develops LED-curable methacrylate composites for rapid fabrication of fiberglass-reinforced materials. The system eliminates thermal and UV curing, reducing energy use and processing time. Mechanical testing confirms performance, supporting efficient and scalable composite manufacturing.	GRAD
181	Wave–Defect Interactions in PAUT and TFM Inspection of Planar Weld Cracks **Manufacturing Engineering**	This study investigates wave–defect interactions in planar weld cracks using PAUT and TFM through combined experimental inspection and validated ultrasonic simulation. Results show TFM provides clearer responses and greater sensitivity to small planar defects than PAUT, while root cracks remain difficult due to weaker reflections and limited tip-diffraction. Findings offer physics-informed guidance for improving detection and characterization of safety-critical weld cracks.	GRAD
182	Microstructural Characterization of Commercial Drywalls under Heat Exposure **Mechanical Engineering**	This research analyzes microstructure and composition of drywall variants before and after high-temperature exposure. SEM-EDX results reveal dehydration, increased porosity, and structural degradation. Findings link microstructural changes to thermal performance, supporting improved fire-resistant material design.	GRAD
183	Hot Deformation Behavior of WAAM 316L Stainless Steel **Mechanical Engineering**	This study evaluates hot deformation behavior of WAAM-produced 316L stainless steel. Results show strong dependence on temperature and strain rate, with improved workability compared to wrought material. Microstructural analysis reveals dynamic recrystallization, supporting improved processing for additive manufacturing.	GRAD
184	Bio-Inspired Aerodynamic Analysis Using Wood Duck Feathers **Mechanical Engineering**	This study evaluates a bio-inspired airfoil using wood duck feathers to improve aerodynamic efficiency. Wind tunnel tests show reduced drag and increased efficiency at certain angles of attack. Results highlight potential applications for UAV and small aircraft design.	GRAD

185	Dirac-like Cone Induced Topologically Protected Wave Trapping in Acoustic Metamaterials **Mechanical Engineering**	This research investigates topologically protected wave propagation in solid-fluid acoustic metamaterials. Results show geometry-driven Dirac-like cone behavior enables energy trapping independent of material composition. Findings support lightweight designs for vibration control and waveguiding applications.	GRAD
186	Machine Learning-Based Defect Identification using Resonant Ultrasonic Spectroscopy **Manufacturing Engineering**	This study applies machine learning to classify defects using resonant ultrasonic spectroscopy data. Frequency signals are processed and analyzed with classification algorithms to distinguish defective from non-defective samples. Results demonstrate improved non-destructive evaluation capabilities.	GRAD
187	Synthesis and Processing of UHMWPE Reinforced with Hybrid Polymer-Coated Nanoparticles **Mechanical Engineering**	This research develops UHMWPE nanocomposite fibers reinforced with polymer-coated CNTs. Results show significant improvements in strength, toughness, and mechanical performance. The study demonstrates advanced processing techniques for high-performance materials in structural applications.	GRAD
Power & Energy Systems (190-209)			
190	Investigating the Application of Thermoelectric Effects in Utilizing Wasted Heat from Dwellings in Georgia	This project investigates thermoelectric generation as a method for converting waste heat from dwellings into electricity. Using a Seebeck-effect apparatus, relationships among temperature, thermocouple voltage, resistance, and thermistor response were characterized. Results support the feasibility of waste-heat recovery and motivate future work with	UG

	as a Form of Renewable Energy **Mechanical Engineering**	higher-performance thermoelectric materials such as bismuth telluride and calcium-manganese oxide to increase voltage output and renewable-energy potential.	
191	Investigation of Renewable HEFA as a Drop-In Fuel for Cleaner Combustion in IC Engines and Civil Aviation **Mechanical Engineering**	This study evaluates Hydroprocessed Esters and Fatty Acids (HEFA) as a renewable drop-in fuel using an instrumented single-cylinder research engine. Compared with ultra-low-sulfur diesel, HEFA showed reduced soot, lower ringing intensity, and improved combustion behavior, while exhaust gas recirculation further reduced NOx. Results support HEFA as a viable lower-emission fuel option for internal combustion engines and civil aviation applications.	UG
192	Preliminary Investigation into the Numerical and Experimental Analysis of Jet-A in a Single Stage Turbojet Engine **Mechanical Engineering**	This research combines experimental measurements and CFD modeling to analyze combustion and emissions of Jet-A fuel in a single-stage turbojet engine. Sensor data for temperature, pressure, thrust, RPM, and fuel flow were used to validate the numerical model. The framework provides a scalable tool for evaluating turbine combustion behavior and supports future studies of sustainable aviation fuels with greater confidence and accuracy.	UG
193	Investigation of Ohnesorge Number's Influence on Primary Spray Atomization of Sustainable Aviation Fuel Blends Analyzed Using Mie Scattering Laser **Mechanical Engineering**	This study examines how changes in the Ohnesorge number affect spray atomization behavior of sustainable aviation fuel blends. Using Mie scattering laser diagnostics, it evaluates droplet distribution and Sauter mean diameter under fixed injection conditions. The work links thermophysical fuel properties to atomization quality, supporting optimization of fuel blends for improved turbine combustion efficiency and reduced aviation emissions.	UG
194	Off-Grid Flywheel-Based Power System **Electrical & Computer Engineering**	This project develops an off-grid flywheel-based power system that converts hand-cranked mechanical energy into regulated DC electricity. The design integrates mechanical storage, power electronics, protection circuitry, and microcontroller-based monitoring to provide stable charging for batteries and low-power loads. The prototype demonstrates a practical emergency power	UG

		solution for remote locations with limited electrical infrastructure.	
195	Efficiency Improvement of the Photovoltaic & Thermal Mobile Solar Powered Water Heater System **Mechanical Engineering**	This project improves a mobile solar-powered water heater by optimizing thermal transfer, fluid circulation, and system reliability. The work addresses overheating, pump performance, plumbing leaks, and collector inefficiencies to enhance sustained operation during long sun exposure. Results support a more durable and efficient photovoltaic-thermal water heating system for sustainable engineering applications.	UG
196	Design, Modeling, and Integration of a Port Fuel Injection System for a Carbureted 35 HP Briggs and Stratton Vanguard V-Twin Engine **Mechanical Engineering**	This project designs and integrates a port fuel injection system for a previously carbureted V-twin utility engine. A new intake manifold with shared plenum and cylinder-specific injectors was modeled and simulated to ensure stable airflow and pressure signals. The work demonstrates a structured method for converting small engines to electronically controlled fuel delivery for improved adaptability and performance.	UG
197	Investigating the Impact of Levee Angle on Voltage Generation in Piezoelectric Wave Energy Harvesters **Mechanical Engineering**	This study investigates how oscillating water column levee angle affects pressure amplification and voltage generation in piezoelectric wave energy harvesters. Transparent models with varying geometries were tested in a wave tank while chamber pressure and PZT output were measured. Results are expected to identify geometry-driven improvements that reduce hydrodynamic losses and increase power generation in turbine-free wave energy systems.	UG
198	A Novel Self-regulating Seal for sCO2 Turbomachinery **Mechanical Engineering**	This project proposes a self-regulating face seal for supercritical CO2 turbomachinery operating under extreme pressure and temperature. The seal tightens as pressure rises while maintaining an sCO2 fluid film, reducing leakage and efficiency losses. Simulations validate the concept, and ongoing studies aim to demonstrate performance for scalable deployment in advanced energy conversion systems.	UG

199	NovaShade: A Supercapacitor-Assisted Windshield Heating System with Closed-Loop Power Control **Electrical & Computer Engineering**	This paper presents NovaShade, an embedded windshield heating system that uses closed-loop temperature control to prevent frost and ice accumulation without idling the engine. The system integrates sensors, PWM-based heating control, voltage protection, and scheduled operation through a companion app. Testing confirmed stable thermal regulation, demonstrating a deployable, fuel-saving smart defroster for vehicle safety.	UG
200	Kalman-Augmented Estimation for Enhancing the Reliability Analysis of Power Electronic Converters Via Sensor-Error Mitigation **Electrical & Computer Engineering**	This research evaluates how sensor errors affect reliability analysis of a buck converter and proposes an augmented Kalman-based estimator to correct monitoring inaccuracies. Simulations show that uncorrected bias, drift, and gain errors distort temperature and lifetime predictions, while the augmented estimator restores values close to the ideal baseline. The approach enables practical, physics-consistent reliability monitoring in sensor-degraded power electronics.	GRAD
201	Experimental Mapping of Power Coefficient and Tip-Speed Ratio for a Vertical-Axis Wind Turbine	This study experimentally maps power coefficient versus tip-speed ratio for a small vertical-axis wind turbine under controlled electrical loading. Using encoder, voltage, current, and wind speed measurements, the work identifies operating conditions for maximum power extraction. Results provide practical performance data for low-power distributed wind applications and clarify how electrical load influences turbine efficiency.	GRAD
202	Optimal Control Framework for Battery Controls in Modern Community Microgrids **Electrical & Computer Engineering**	This study proposes an optimal control framework for battery current allocation in microgrids to reduce aging and extend service life. The method continuously adjusts control parameters using real-time operating conditions, protecting weaker or temperature-stressed battery packs while maintaining stability. Results indicate low aging risk, robust performance, and economically efficient long-term operation for multi-battery community microgrids.	GRAD

Simulation & Modeling (210-229)			
210	Aerodynamic Performance Improvement Analysis of Aircraft Wings **Mechanical Engineering**	This study investigates aerodynamic performance of aircraft wings by analyzing lift, drag, and flow behavior under varying conditions. Using simplified models, it evaluates parameters such as Reynolds number, angle of attack, and wing geometry. Results highlight how camber and aspect ratio influence lift-to-drag ratio, providing insight into optimizing wing design for improved aerodynamic efficiency and early-stage aircraft development.	UG
211	Convexification of Optimal Isotropic Sensor Placement for Barrier Coverage Systems **Electrical & Computer Engineering**	This research develops a convex optimization formulation for sensor placement in barrier coverage systems. By overcoming limitations of nonconvex mixed-integer approaches, the method aims to achieve globally optimal solutions for detection probability. The study compares solution performance and computational efficiency, supporting improved real-time and offline deployment of sensor networks.	UG
212	Computational Aeroelastic Analysis of Wings in High-Mach-Number Flow Regimes **Mechanical Engineering**	This study examines aeroelastic behavior of wings in supersonic flow, focusing on interactions among aerodynamic forces, structural dynamics, and inertial effects. Using computational and analytical models, it evaluates phenomena such as flutter and shock-induced oscillations. The work supports improved design and stability of high-speed aerospace systems.	UG
220	Stress-Limiter Sensitivity and Bifurcation Structure of Transonic Shock Buffet **Mechanical Engineering**	This research analyzes sensitivity of transonic shock buffet predictions to turbulence model parameters. A parametric study identifies bifurcation behavior and transition thresholds between steady and oscillatory flow. Results improve accuracy of computational models and provide guidance for predicting buffet onset in aerodynamic design.	GRAD
221	Thermal Decomposition of Aerospace	This study develops physics-based models to simulate thermal decomposition of aerospace composites under fire conditions. Coupling heat transfer, gas transport, and chemical reactions, the	GRAD

	Composite Materials under Fire Exposure **Mechanical Engineering**	model predicts ignition behavior and material degradation. Results support improved fire safety analysis and certification of composite materials.	
222	Thermal Decomposition Modeling and Validation for Aerospace Carbon/Epoxy Materials **Mechanical Engineering**	This research applies and validates the ThermaKin2D pyrolysis model for carbon fiber composites under fire exposure. Model predictions are compared with experimental data, demonstrating accurate representation of decomposition behavior. The work supports broader application of predictive models for aerospace material safety.	GRAD
223	One-Dimensional Neuron Modeling for Non-invasive Neural Testing **Mechanical Engineering**	This study develops a one-dimensional computational neuron model to analyze electrochemical behavior under varying membrane conditions. Using MATLAB, the model simulates neural responses to changes in voltage and external stimuli, supporting research in neuromodulation and treatment of neurological disorders.	GRAD
224	Aeroacoustic Performance Enhancement of Airfoils Using Passive Flow Control **Mechanical Engineering**	This research investigates passive flow control using surface dimples to reduce trailing-edge noise in airfoils. CFD and aeroacoustic simulations evaluate effects on turbulence and sound generation. Results aim to reduce broadband noise while maintaining aerodynamic performance.	GRAD
225	Effect of Leading Edge Vortex Generators on Aerodynamics of a Fixed Wing **Mechanical Engineering**	This study evaluates leading-edge vortex generators for improving aerodynamic performance. CFD simulations and wind tunnel experiments analyze lift, drag, and flow separation. Results indicate delayed stall and improved efficiency, supporting passive flow control applications in aerospace and wind energy.	GRAD
226	Real-Time Pallet Label Verification Using OCR and WMS Integration in Warehouse Logistics	This study presents a real-time framework integrating OCR with Warehouse Management System (WMS) validation for automated pallet label identification. Using a telepresence robot and cloud-based OCR, extracted part numbers are verified against a	GRAD

	Computer Science	database to detect mismatches. Experiments across varying fonts, distances, and angles show robust OCR performance once minimum thresholds are met. Results demonstrate that database validation is essential alongside OCR, providing a scalable solution for warehouse logistics automation.	